MATHEMATICS BOOK

Test Preparation - Notes & MCQs

PST / JEST - SPSC - Screening Tests

Matric Level - Pakistan Test Pattern

Short Notes - Worked Examples - 200 Practice MCQs with Answer Keys

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How to use this book: Har chapter mein pehle short notes parhein, phir worked examples dekhein, aur aakhir mein practice MCQs khud hal karein. Jawab chapter ke aakhir mein diye gaye Answer Key se check karein. Roz ek chapter parhna sab se behtareen tareeqa hai. Galat honey wale sawal dobara zaroor karein - galti hi sab se achi teacher hoti hai.

Chapter 1

Number System & Basic Operations

Har test ki bunyaad numbers hain. Is chapter mein hum number ki types, prime numbers, HCF/LCM aur basic hisaab (operations) seekhenge.

Notes - Asaan Alfaaz Mein

- **Natural numbers:** 1, 2, 3, 4, ... (ginne wale numbers). **Whole numbers** mein 0 bhi shamil ho jata hai: 0, 1, 2, 3, ...
- **Integers:** ..., -3, -2, -1, 0, 1, 2, 3, ... (negative numbers bhi shamil hain).
- **Even numbers** 2 par poora divide hote hain (2, 4, 6, ...). **Odd numbers** 2 par divide nahi hote (1, 3, 5, ...).
- **Prime number:** woh number jiske sirf do factors hon - 1 aur woh khud. Misal: 2, 3, 5, 7, 11, 13. Yaad rakhein: 1 na prime hai na composite. 2 sab se chhota prime hai aur akela even prime hai.
- **Composite numbers:** jin ke do se zyada factors hon, jaise 4, 6, 8, 9, 10.
- **HCF (Highest Common Factor):** do numbers ka sab se bara common factor. **LCM (Least Common Multiple):** do numbers ka sab se chhota common multiple.
- **Divisibility trick:** agar digits ka sum 3 par divide ho jaye to number 3 par divide hota hai. Digits ka sum 9 ho to number 9 par bhi divide hota hai.
- **BODMAS rule:** pehle Brackets, phir Orders (powers), phir Division, Multiplication, Addition, Subtraction - is order mein sawal hal karein.
- **Absolute value:** |-7| = 7. Number line par 0 se doori hamesha positive hoti hai.

Worked Examples

Example 1: HCF aur LCM nikalein - 12 aur 18

12 ke factors: 1, 2, 3, 4, 6, 12

18 ke factors: 1, 2, 3, 6, 9, 18

Common factors: 1, 2, 3, 6 - sab se bara = 6, to HCF = 6.

12 ke multiples: 12, 24, 36, ... aur 18 ke multiples: 18, 36, ... - sab se chhota common = 36, to LCM = 36.

Example 2: BODMAS ka sawal: 8 + 4 x 3

Pehle multiplication: 4 x 3 = 12

Phir addition: 8 + 12 = 20

Jawab = 20 (agar pehle 8 + 4 kar dein to 36 aayega, jo ghalat hai).

Example 3: LCM of 6 and 8

6 ke multiples: 6, 12, 18, 24, ...

8 ke multiples: 8, 16, 24, ...

Pehla common multiple = 24, to LCM = 24.

Aam Ghaltiyan - Common Mistakes

- BODMAS bhool kar sawal ko left se right hal karna - pehle hamesha multiply/divide karein.
- 1 ko prime number samajh lena - 1 na prime hai na composite.

Practice MCQs

1. Which of the following is a prime number?

(a) 29 (b) 21 (c) 27 (d) 33

2. The smallest even prime number is:

(a) 1 (b) 2 (c) 3 (d) 4

3. What is the HCF of 12 and 18?

(a) 3 (b) 4 (c) 6 (d) 9

4. What is the LCM of 4 and 6?

(a) 24 (b) 8 (c) 10 (d) 12

5. Which number is divisible by 3?

(a) 124 (b) 311 (c) 231 (d) 415

6. 45 + 37 = ?

(a) 82 (b) 72 (c) 92 (d) 78

7. 96 / 8 = ?

(a) 11 (b) 13 (c) 14 (d) 12

8. 15 x 6 = ?

(a) 80 (b) 90 (c) 85 (d) 95

9. The square of 13 is:

(a) 156 (b) 169 (c) 182 (d) 144

10. The cube of 4 is:

(a) 16 (b) 32 (c) 48 (d) 64

11. The value of |-7| is:

(a) -7 (b) 0 (c) 7 (d) 14

12. 25 - (-5) = ?

(a) 30 (b) 20 (c) -20 (d) -30

13. Which of the following is an odd number?

(a) 248 (b) 360 (c) 802 (d) 517

14. What is the place value of 7 in 5,372?

(a) 7 (b) 70 (c) 700 (d) 7000

15. 8 + 4 x 3 = ?

(a) 20 (b) 36 (c) 24 (d) 16

16. The smallest 3-digit number is:

(a) 99 (b) 101 (c) 100 (d) 111

17. 1000 - 457 = ?

(a) 543 (b) 553 (c) 443 (d) 533

Answer Key - Chapter 1

1. (a) 2. (b) 3. (c) 4. (d) 5. (c) 6. (a) 7. (d) 8. (b) 9. (b) 10. (d) 11. (c) 12. (a) 13. (d) 14. (b) 15. (a) 16. (c) 17. (a)

Try Yourself - Quick Drills

1. 37 + 48 = ?
2. 13 x 5 = ?
3. HCF of 8 and 12 = ?
4. 7 ke baad agla prime number?
5. 250 - 97 = ?

Drill Answers: 1. 85 2. 65 3. 4 4. 11 5. 153

Chapter 2

Fractions & Decimals

Fraction kisi cheez ka hissa (part) zahir karta hai. Tests mein fractions aur decimals ke sawal har saal aate hain - ye wala chapter mazboot kar lein.

Notes - Asaan Alfaaz Mein

- **Fraction** a/b ki surat mein likha jata hai. Upar wala hissa **numerator**, neeche wala **denominator** kehlata hai.
- **Proper fraction:** numerator chhota ho (3/5). **Improper fraction:** numerator bara ho (7/4). **Mixed number:** 2 1/3 jaisa number - poora number + fraction.
- Mixed number ko improper banane ke liye: 2 1/3 = (2 x 3 + 1)/3 = 7/3.
- **Fractions ka addition/subtraction:** pehle denominators ek jaise (LCM se) karein, phir numerators jama ya minus karein.
- **Multiplication:** numerator x numerator, denominator x denominator. **Division:** dusri fraction ko ulta (reciprocal) karke multiply karein.
- **Reciprocal:** 4/7 ka reciprocal 7/4 hai - bas ulta kar dein.
- **Decimals:** 0.75 = 75/100 = 3/4. Fraction ko decimal banane ke liye numerator ko denominator se divide karein.
- Decimals jama/minus karte waqt decimal point ko seedha (align) rakhein: 4.60 + 2.35 = 6.95.

Worked Examples

Example 1: 1/2 + 1/4 hal karein

Denominators 2 aur 4 ka LCM = 4.

1/2 = 2/4, to sawal bana 2/4 + 1/4 = 3/4.

Jawab = 3/4.

Example 2: 3/4 / 1/2 hal karein

Division mein dusri fraction ulta ho jati hai: 3/4 x 2/1

= 6/4 = 3/2.

Jawab = 3/2.

Example 3: 0.4 + 3/5 hal karein

Pehle 3/5 ko decimal banayein: 3/5 = 0.6

Ab 0.4 + 0.6 = 1.0

Jawab = 1.0 (fraction aur decimal mila kar hal karna asaan hota hai).

Aam Ghaltiyan - Common Mistakes

- Fractions jama karte waqt denominators ko bhi jama kar dena (1/2 + 1/3 ko 2/5 likhna ghalat hai).
- Decimals jama karte waqt decimal points ko seedha (align) na karna.

Practice MCQs

1. 1/2 + 1/4 = ?

(a) 2/6 (b) 1/6 (c) 3/4 (d) 5/4

2. Which fraction is the largest?

(a) 2/3 (b) 3/4 (c) 5/8 (d) 7/12

3. 5/6 - 1/3 = ?

(a) 2/3 (b) 1/3 (c) 4/9 (d) 1/2

4. 2/3 x 3/5 = ?

(a) 6/15 (b) 5/6 (c) 1/5 (d) 2/5

5. 3/4 / 1/2 = ?

(a) 3/2 (b) 3/8 (c) 2/3 (d) 1/4

6. The mixed number 2 1/3 as an improper fraction is:

(a) 5/3 (b) 6/3 (c) 7/3 (d) 7/2

7. 0.75 written as a fraction in lowest form is:

(a) 7/10 (b) 3/4 (c) 4/5 (d) 2/3

8. 3/5 written as a decimal is:

(a) 0.35 (b) 0.6 (c) 0.06 (d) 0.65

9. 4.6 + 2.35 = ?

(a) 6.85 (b) 7.05 (c) 6.95 (d) 6.90

10. 7.5 x 4 = ?

(a) 28 (b) 32 (c) 30.5 (d) 30

11. Which decimal is the smallest?

(a) 0.205 (b) 0.25 (c) 0.52 (d) 0.5

12. 5/8 of 64 is:

(a) 32 (b) 45 (c) 40 (d) 48

13. Which fraction is equal to 2/3?

(a) 6/12 (b) 4/9 (c) 8/9 (d) 8/12

14. 1.25 + 0.75 = ?

(a) 2.0 (b) 1.90 (c) 2.5 (d) 1.10

15. The reciprocal of 4/7 is:

(a) 4/7 (b) 7/4 (c) 1/4 (d) 3/4

16. 9.03 - 4.5 = ?

(a) 4.48 (b) 5.47 (c) 4.35 (d) 4.53

17. 7/10 written as a percent is:

(a) 7% (b) 17% (c) 70% (d) 700%

Answer Key - Chapter 2

1. (c) 2. (b) 3. (d) 4. (d) 5. (a) 6. (c) 7. (b) 8. (b) 9. (c) 10. (d) 11. (a) 12. (c) 13. (d) 14. (a) 15. (b) 16. (d) 17. (c)

Try Yourself - Quick Drills

1. 1/3 + 1/3 = ?
2. 0.9 as a fraction = ?
3. 3/4 of 20 = ?
4. 2.5 x 2 = ?
5. 6/8 in lowest form = ?

Drill Answers: 1. 2/3 2. 9/10 3. 15 4. 5 5. 3/4

Chapter 3

Percentages

Percent ka matlab hai "har 100 mein se". Percentage ke sawal har screening test mein lazmi aate hain - marks, salary, qeemat, sab jagah yehi formula chalta hai.

Notes - Asaan Alfaaz Mein

- **Percent to fraction:** 25% = 25/100 = 1/4. **Percent to decimal:** 25% = 0.25.
- Kuch mashhoor values yaad kar lein: 50% = 1/2, 25% = 1/4, 75% = 3/4, 20% = 1/5, 10% = 1/10, 33.33% = 1/3.
- **Percent of a quantity:** (Percent / 100) x Quantity. Misal: 20% of 150 = 20/100 x 150 = 30.
- **Percentage nikalna:** (Part / Whole) x 100. Misal: 45 out of 50 = 45/50 x 100 = 90%.
- **Increase:** New value = Old + (Percent/100 x Old). **Decrease:** New value = Old - (Percent/100 x Old).
- **Percentage change:** (Change / Original) x 100 - hamesha ORIGINAL (purani) value se divide karein, ye aam ghalti hai.

Worked Examples

Example 1: 20% of 150 nikalein

20% = 20/100 = 1/5

150 ka 1/5 = 150 / 5 = 30.

Jawab = 30.

Example 2: Ek student ne 60 mein se 48 numbers liye - percentage?

Percentage = 48/60 x 100

= 0.8 x 100 = 80%.

Example 3: Qeemat Rs 200 se Rs 250 ho gayi - kitne percent izafa?

Change = 250 - 200 = 50

Percentage increase = 50/200 x 100 = 25%.

Example 4: 150 ka 30% nikalein

Pehle 10% nikalein: 150 ka 10% = 15

30% = 3 x 10% = 3 x 15 = 45.

Aam Ghaltiyan - Common Mistakes

- Percentage change mein new (bari hui) value se divide karna - hamesha original value se divide karein.
- 0.5 ko 5% samajhna - asal mein 0.5 = 50% hota hai.

Practice MCQs

1. 50% of 240 is:

(a) 100 (b) 120 (c) 140 (d) 60

2. 25% of 80 is:

(a) 20 (b) 25 (c) 40 (d) 15

3. 10% of 450 is:

(a) 40 (b) 45 (c) 90 (d) 35

4. 0.35 written as a percent is:

(a) 35% (b) 3.5% (c) 0.35% (d) 350%

5. 3/4 written as a percent is:

(a) 34% (b) 43% (c) 7.5% (d) 75%

6. 20% of a number is 60. The number is:

(a) 120 (b) 200 (c) 300 (d) 240

7. A student scored 45 marks out of 50. His percentage is:

(a) 90% (b) 80% (c) 85% (d) 95%

8. A price rises from Rs 200 to Rs 250. The percentage increase is:

(a) 20% (b) 30% (c) 50% (d) 25%

9. 15% of 300 is:

(a) 30 (b) 60 (c) 45 (d) 35

10. In a class of 50 students, 60% are boys. The number of boys is:

(a) 20 (b) 30 (c) 35 (d) 40

11. Decrease 80 by 25%. The result is:

(a) 55 (b) 65 (c) 60 (d) 75

12. What percent of 200 is 50?

(a) 20% (b) 25% (c) 40% (d) 50%

13. A salary of Rs 40,000 is increased by 10%. The new salary is:

(a) Rs 44,000 (b) Rs 40,400 (c) Rs 41,000 (d) Rs 48,000

14. 5% of 1000 is:

(a) 5 (b) 500 (c) 105 (d) 50

15. A shirt of Rs 800 is sold at 20% discount. The buyer pays:

(a) Rs 640 (b) Rs 600 (c) Rs 700 (d) Rs 780

16. A score improves from 40 to 50. The percentage increase is:

(a) 10% (b) 25% (c) 20% (d) 30%

17. One third (33.33%) of 90 is:

(a) 33 (b) 45 (c) 30 (d) 60

18. If a number is increased by 50%, the new number is:

(a) 2 times the old (b) 0.5 times the old (c) 50 times the old (d) 1.5 times the old

Answer Key - Chapter 3

1. (b) 2. (a) 3. (b) 4. (a) 5. (d) 6. (c) 7. (a) 8. (d) 9. (c) 10. (b) 11. (c) 12. (b) 13. (a) 14. (d) 15. (a) 16. (b) 17. (c) 18. (d)

Try Yourself - Quick Drills

1. 50% of 90 = ?
2. 1/5 as a percent = ?
3. 10% of 70 = ?
4. 5% of 200 = ?
5. 75% of 40 = ?

Drill Answers: 1. 45 2. 20% 3. 7 4. 10 5. 30

Chapter 4

Ratio & Proportion

Ratio do cheezon ka muqabla (comparison) hai. Khana taqseem ho, naqsha (map) banana ho ya qeemat nikalna ho - ratio har jagah kaam aata hai.

Notes - Asaan Alfaaz Mein

- **Ratio** a : b ki surat mein likha jata hai. Isko hamesha sab se asaan (lowest) form mein likhte hain: 12 : 18 = 2 : 3.
- Units ek jaisi karna lazmi hai: 500 g : 2 kg pehle 500 g : 2000 g = 1 : 4 banega.
- **Amount ko ratio mein taqseem karna:** pehle ratio ke parts ka total nikalein, phir har hisse ki value = (uska part / total parts) x total amount.
- **Proportion:** do ratios ki barabari. 2 : 3 = 4 : 6 ek proportion hai. Ismein cross-multiplication chalta hai: 2 x 6 = 3 x 4 = 12.
- **Direct proportion:** ek cheez barhe to dusri bhi barhe (zyada pens, zyada qeemat). Pehle ek cheez (unit) ki value nikalein.
- **Inverse proportion:** ek barhe to dusri ghate (zyada mazdoor, kam din). Total kaam (men x days) barabar rehta hai.

Worked Examples

Example 1: Rs 600 ko 2 : 3 ke ratio mein taqseem karein

Total parts = 2 + 3 = 5. Ek part = 600 / 5 = Rs 120.

Pehla hissa = 2 x 120 = Rs 240. Dusra hissa = 3 x 120 = Rs 360.

Example 2: 3 pens ki qeemat Rs 60 hai - 7 pens ki qeemat?

Ek pen = 60 / 3 = Rs 20.

7 pens = 7 x 20 = Rs 140.

Example 3: Rs 900 ko 4 : 5 ke ratio mein taqseem karein

Total parts = 4 + 5 = 9. Ek part = 900 / 9 = Rs 100.

Hisay: 4 x 100 = Rs 400 aur 5 x 100 = Rs 500.

Aam Ghaltiyan - Common Mistakes

- Alag units ko baghair badle ratio banana (kg aur g ko seedha ratio mein likh dena).
- Ratio ko ulta likh dena - a : b aur b : a mein bohat farq hai.

Practice MCQs

1. The ratio 12 : 18 in lowest form is:

(a) 3 : 2 (b) 4 : 5 (c) 2 : 3 (d) 3 : 5

2. The ratio of 500 g to 2 kg is:

(a) 1 : 4 (b) 1 : 2 (c) 4 : 1 (d) 2 : 5

3. Rs 600 is divided in the ratio 2 : 3. The larger share is:

(a) Rs 240 (b) Rs 300 (c) Rs 400 (d) Rs 360

4. If a : b = 3 : 5 and a = 15, then b = ?

(a) 9 (b) 25 (c) 20 (d) 30

5. Which of the following is a proportion?

(a) 2 : 3 = 3 : 4 (b) 2 : 3 = 4 : 6 (c) 5 : 2 = 10 : 5 (d) 1 : 4 = 2 : 6

6. If 3 pens cost Rs 60, then 7 pens cost:

(a) Rs 120 (b) Rs 160 (c) Rs 105 (d) Rs 140

7. On a map, 1 cm shows 5 km. A distance of 4 cm on the map is:

(a) 9 km (b) 15 km (c) 20 km (d) 25 km

8. The ratio of 45 minutes to 1 hour is:

(a) 3 : 4 (b) 4 : 3 (c) 3 : 5 (d) 5 : 4

9. If x : 4 = 9 : 6, then x = ?

(a) 3 (b) 12 (c) 24 (d) 6

10. Two numbers are in the ratio 4 : 7 and their sum is 55. The smaller number is:

(a) 28 (b) 20 (c) 35 (d) 25

11. If 8 books cost Rs 400, the cost of one book is:

(a) Rs 50 (b) Rs 40 (c) Rs 45 (d) Rs 55

12. A car covers 150 km in 3 hours at the same speed. In 5 hours it will cover:

(a) 200 km (b) 225 km (c) 250 km (d) 300 km

13. Boys and girls are in the ratio 5 : 3 in a class of 40. The number of girls is:

(a) 15 (b) 25 (c) 20 (d) 12

14. The ratio 15 : 25 in lowest form is:

(a) 5 : 3 (b) 1 : 5 (c) 3 : 5 (d) 3 : 4

15. If a/2 = b/3 and a + b = 25, then a = ?

(a) 15 (b) 10 (c) 5 (d) 20

16. 6 men can finish a work in 10 days. In how many days will 12 men finish the same work?

(a) 20 days (b) 8 days (c) 15 days (d) 5 days

17. The fourth proportional to 2, 4 and 6 is:

(a) 3 (b) 8 (c) 10 (d) 12

Answer Key - Chapter 4

1. (c) 2. (a) 3. (d) 4. (b) 5. (b) 6. (d) 7. (c) 8. (a) 9. (d) 10. (b) 11. (a) 12. (c) 13. (a) 14. (c) 15. (b) 16. (d) 17. (d)

Try Yourself - Quick Drills

1. 6 : 9 in lowest form = ?

2. 1 : 2 = x : 10, to x = ?

3. Rs 100 ko 1 : 4 mein taqseem karein (chhota hissa)

4. 30 minutes : 1 hour ka ratio = ?

5. 2 kg aalu Rs 300 ke hain - 1 kg = ?

Drill Answers: 1. 2 : 3 2. 5 3. Rs 20 4. 1 : 2 5. Rs 150

Chapter 5

Average

Average (ost) roz-marra ki cheez hai - average marks, average age, average speed. Formula ek hi hai, sawal ki surat badal jati hai.

Notes - Asaan Alfaaz Mein

- **Average = (Sum of all values) / (Number of values).** Yehi ek formula poora chapter hai.
- Agar average aur tadaad maloom ho to **Sum = Average x Number of values**.
- Ghaib (missing) number nikalne ke liye: pehle total sum nikalein, phir maloom numbers minus kar dein.
- **Shortcut:** agar numbers barabar fasle par hon (7, 14, 21, 28, 35) to average hamesha darmiyan (middle) wala number hota hai - yahan 21.
- Average speed ke sawalon mein hamesha total distance ko total time se divide karein.

Worked Examples

Example 1: 10, 20, 30 aur 40 ka average nikalein

Sum = 10 + 20 + 30 + 40 = 100

Values ki tadaad = 4

Average = 100 / 4 = 25.

Example 2: 4 numbers ka average 20 hai. Teen numbers 15, 20, 25 hain - chautha number?

Total sum = 20 x 4 = 80

Maloom numbers ka sum = 15 + 20 + 25 = 60

Chautha number = 80 - 60 = 20.

Example 3: 2, 5 aur 8 ka average

Sum = 2 + 5 + 8 = 15

Average = 15 / 3 = 5.

Aam Ghaltiyan - Common Mistakes

- Average nikalte waqt values ki tadaad galat ginna.
- Average speed ke liye do speeds ka seedha average le lena.

Practice MCQs

1. The average of 10, 20, 30 and 40 is:

(a) 25 (b) 20 (c) 30 (d) 35

2. The average of the first five natural numbers (1 to 5) is:

(a) 2.5 (b) 4 (c) 3 (d) 15

3. The average of 8 and 12 is:

(a) 9 (b) 10 (c) 11 (d) 20

4. The average of 6 numbers is 15. Their sum is:

(a) 60 (b) 90 (c) 75 (d) 100

5. A batsman scores 40, 50 and 60 runs in 3 matches. His average score is:

(a) 45 (b) 55 (c) 50 (d) 150

6. The average of ages 12, 14 and 16 years is:

(a) 13 years (b) 15 years (c) 42 years (d) 14 years

7. The average of 4 numbers is 20. Three of them are 15, 20 and 25. The fourth number is:

(a) 20 (b) 15 (c) 25 (d) 30

8. The average of 2, 4, 6, 8 and 10 is:

(a) 5 (b) 7 (c) 6 (d) 30

9. The temperatures on five days were 30, 32, 34, 36 and 38 degrees. The average temperature is:

(a) 32 degrees (b) 35 degrees (c) 36 degrees (d) 34 degrees

10. The average of 100 and 200 is:

(a) 150 (b) 120 (c) 140 (d) 160

11. A car travels 60 km in the first hour and 40 km in the second hour. Its average speed is:

(a) 45 km/h (b) 50 km/h (c) 55 km/h (d) 100 km/h

12. If the average of x and 15 is 20, then x = ?

(a) 5 (b) 30 (c) 35 (d) 25

13. The average of 7, 14, 21, 28 and 35 is:

(a) 20 (b) 22 (c) 21 (d) 105

14. The average marks of a class are 72. The total marks of 25 students are:

(a) 720 (b) 1800 (c) 900 (d) 1750

15. The average of the first ten counting numbers (1 to 10) is:

(a) 5.5 (b) 5 (c) 6 (d) 10

16. The average of 0, 10, 20, 30 and 40 is:

(a) 25 (b) 20 (c) 15 (d) 100

17. The average weight of 4 boys is 40 kg. Their total weight is:

(a) 160 kg (b) 120 kg (c) 140 kg (d) 180 kg

Answer Key - Chapter 5

1. (a) 2. (c) 3. (b) 4. (b) 5. (c) 6. (d) 7. (a) 8. (c) 9. (d) 10. (a) 11. (b) 12. (d) 13. (c) 14. (b) 15. (a) 16. (b) 17. (a)

Try Yourself - Quick Drills

1. Average of 4 and 8 = ?
2. Average of 10, 20 and 60 = ?
3. 5 numbers ka average 12 hai - un ka sum = ?
4. Average of 1, 2 and 3 = ?
5. Teen numbers ka average 10 hai. Do numbers 8 aur 12 hain - teesra = ?

Drill Answers: 1. 6 2. 30 3. 60 4. 2 5. 10

Chapter 6

Profit & Loss

Khareed-o-farokht (buying and selling) ke sawal har test mein aate hain. Bas do cheezein yaad rakhein: CP (lagat) aur SP (farokht ki qeemat).

Notes - Asaan Alfaaz Mein

- **CP = Cost Price** (jis qeemat par cheez khareedi), **SP = Selling Price** (jis par bechi). **MP = Marked Price** (cheez par likhi qeemat).
- **Profit = SP - CP** (jab SP zyada ho). **Loss = CP - SP** (jab CP zyada ho).
- **Profit % = (Profit / CP) x 100** aur **Loss % = (Loss / CP) x 100**. Yaad rakhein: percent hamesha CP par nikalta hai, SP par nahi - ye sab se aam ghalti hai.
- **SP nikalna:** SP = CP x (100 + Profit%) / 100. Loss ho to SP = CP x (100 - Loss%) / 100.
- **Discount:** Discount = MP x Discount% / 100, aur SP = MP - Discount.
- **CP nikalna (SP aur % se):** CP = SP x 100 / (100 + Profit%) ya CP = SP x 100 / (100 - Loss%).

Worked Examples

Example 1: Ek cheez Rs 500 mein khareedi aur Rs 600 mein bechi - profit aur profit %?

Profit = 600 - 500 = Rs 100

Profit % = 100/500 x 100 = 20%.

Example 2: CP Rs 400 hai aur loss 5% hai - SP nikalein

Loss = 5% of 400 = Rs 20

SP = 400 - 20 = Rs 380.

Example 3: Marked price Rs 500 par 20% discount - SP?

Discount = 20% of 500 = Rs 100

SP = 500 - 100 = Rs 400.

Aam Ghaltiyan - Common Mistakes

- Profit percent ko SP par nikalna - percent hamesha CP par banta hai.
- Discount ko CP par samajhna - discount hamesha marked price (MP) par milta hai.

Practice MCQs

1. An article is bought for Rs 500 and sold for Rs 600. The profit is:

(a) Rs 50 (b) Rs 150 (c) Rs 1100 (d) Rs 100

2. An article bought for Rs 500 is sold for Rs 600. The profit percent is:

(a) 10% (b) 25% (c) 20% (d) 16%

3. A bag is bought for Rs 800 and sold for Rs 720. The loss is:

(a) Rs 80 (b) Rs 60 (c) Rs 100 (d) Rs 90

4. A bag bought for Rs 800 is sold at a loss of Rs 80. The loss percent is:

(a) 8% (b) 12% (c) 20% (d) 10%

5. A man buys a watch for Rs 200 and sells it for Rs 250. His profit percent is:

(a) 20% (b) 50% (c) 25% (d) 10%

6. The cost price of a chair is Rs 1000. It is sold at 10% profit. The selling price is:

(a) Rs 1010 (b) Rs 1100 (c) Rs 1090 (d) Rs 1110

7. The cost price of a fan is Rs 400. It is sold at 5% loss. The selling price is:

(a) Rs 390 (b) Rs 395 (c) Rs 380 (d) Rs 360

8. An article is bought for Rs 750 and sold for Rs 900. The profit percent is:

(a) 15% (b) 20% (c) 25% (d) 16%

9. A toy is bought for Rs 350 and sold for Rs 350. The profit is:

(a) Rs 0 (b) Rs 350 (c) Rs 700 (d) Rs 35

10. A mobile is bought for Rs 1200 and sold for Rs 1500. The profit percent is:

(a) 20% (b) 30% (c) 15% (d) 25%

11. A jacket has marked price Rs 1000 and is sold at 10% discount. The selling price is:

(a) Rs 900 (b) Rs 910 (c) Rs 990 (d) Rs 800

12. An article is bought for Rs 250 and sold for Rs 200. The loss percent is:

(a) 25% (b) 20% (c) 10% (d) 15%

13. A shopkeeper buys pencils at Rs 10 each and sells them at Rs 12 each. His profit percent is:

(a) 10% (b) 25% (c) 20% (d) 12%

14. A book with cost price Rs 600 is sold at 15% profit. The selling price is:

(a) Rs 615 (b) Rs 660 (c) Rs 715 (d) Rs 690

15. A cycle is bought for Rs 2000 and sold for Rs 1700. The loss percent is:

(a) 10% (b) 20% (c) 15% (d) 25%

16. An item is sold for Rs 540 at a loss of 10%. Its cost price is:

(a) Rs 600 (b) Rs 594 (c) Rs 480 (d) Rs 550

17. Five shirts are bought at Rs 700 each and sold at Rs 800 each. The total profit is:

(a) Rs 100 (b) Rs 400 (c) Rs 700 (d) Rs 500

Answer Key - Chapter 6

1. (d) 2. (c) 3. (a) 4. (d) 5. (c) 6. (b) 7. (c) 8. (b) 9. (a) 10. (d) 11. (a) 12. (b) 13. (c) 14. (d) 15. (c) 16. (a) 17. (d)

Try Yourself - Quick Drills

1. CP Rs 100, SP Rs 130 - profit % = ?

2. CP Rs 500, loss Rs 50 - SP = ?

3. CP Rs 200, SP Rs 260 - profit = ?

4. MP Rs 200 par 10% discount - SP = ?

5. SP Rs 220, profit 10% - CP = ?

Drill Answers: 1. 30% 2. Rs 450 3. Rs 60 4. Rs 180 5. Rs 200

Chapter 7

Simple Interest

Bank mein rakhi rakam par jo extra paisa milta hai use sood (interest) kehte hain. Simple interest mein har saal sood sirf asal rakam (principal) par banta hai.

Notes - Asaan Alfaaz Mein

- **P** = Principal (asal rakam), **R** = Rate (% per year), **T** = Time (years), **SI** = Simple Interest.
- **SI = (P x R x T) / 100** - ye ek formula poora chapter hal kar deta hai.
- **Amount (kul rakam) = P + SI.**
- Formula ko ulta bhi chala sakte hain: P = SI x 100 / (R x T), R = SI x 100 / (P x T), T = SI x 100 / (P x R).
- Agar time months mein ho to years mein badlein: 6 months = 1/2 year, 3 months = 1/4 year.
- Hint: SI hamesha P, R aur T teenon ke barhne se barhta hai - jawab check karne ka asaan tareeqa.

Worked Examples

Example 1: Rs 1000 par 5% per year ke hisab se 2 saal ka simple interest

SI = (P x R x T) / 100 = (1000 x 5 x 2) / 100

= 10000 / 100 = Rs 100.

Example 2: SI Rs 300 hai, P Rs 1500 aur R 10% hai - time nikalein

T = SI x 100 / (P x R) = 300 x 100 / (1500 x 10)

= 30000 / 15000 = 2 years.

Example 3: Rs 3000 par 5% se 2 saal - amount nikalein

SI = (3000 x 5 x 2) / 100 = Rs 300

Amount = 3000 + 300 = Rs 3300.

Aam Ghaltiyan - Common Mistakes

- Months ko years mein badle baghair formula lagana (6 months = 1/2 year).
- Amount aur SI ko ek hi cheez samajhna - Amount = P + SI hota hai.

Practice MCQs

1. The simple interest on Rs 1000 at 5% per year for 2 years is:

(a) Rs 50 (b) Rs 100 (c) Rs 150 (d) Rs 200

2. The simple interest on Rs 2000 at 10% per year for 3 years is:

(a) Rs 300 (b) Rs 600 (c) Rs 500 (d) Rs 700

3. Rs 500 is kept at 4% per year for 2 years. The total amount after 2 years is:

(a) Rs 520 (b) Rs 560 (c) Rs 580 (d) Rs 540

4. The simple interest on Rs 1000 in 4 years is Rs 200. The rate of interest is:

(a) 4% (b) 10% (c) 5% (d) 2%

5. On Rs 1500 at 10% per year, the simple interest of Rs 300 is earned in:

(a) 2 years (b) 1 year (c) 3 years (d) 4 years

6. The simple interest on Rs 5000 at 6% per year for 1 year is:

(a) Rs 60 (b) Rs 600 (c) Rs 330 (d) Rs 300

7. The simple interest on Rs 800 at 5% per year for 3 years is:

(a) Rs 100 (b) Rs 120 (c) Rs 150 (d) Rs 240

8. The principal is Rs 2000 and the simple interest is Rs 240. The amount is:

(a) Rs 2240 (b) Rs 2040 (c) Rs 2400 (d) Rs 1760

9. The simple interest on Rs 2500 at 8% per year for 2 years is:

(a) Rs 200 (b) Rs 450 (c) Rs 400 (d) Rs 500

10. Using SI = (P x R x T)/100, the interest on Rs 1200 at 3% for 2 years is:

(a) Rs 72 (b) Rs 36 (c) Rs 720 (d) Rs 120

11. The simple interest on Rs 3000 in 3 years is Rs 450. The rate is:

(a) 3% (b) 15% (c) 5% (d) 6%

12. The simple interest on Rs 600 at 12% per year for 5 years is:

(a) Rs 300 (b) Rs 360 (c) Rs 420 (d) Rs 72

13. At 6% per year for 3 years, the simple interest is Rs 180. The principal is:

(a) Rs 900 (b) Rs 1080 (c) Rs 600 (d) Rs 1000

14. Rs 1500 is kept at 4% per year for 2 years. The amount after 2 years is:

(a) Rs 1580 (b) Rs 1600 (c) Rs 1700 (d) Rs 1620

15. The simple interest on Rs 2000 at 10% per year for 6 months is:

(a) Rs 100 (b) Rs 200 (c) Rs 50 (d) Rs 150

16. A bank pays Rs 90 interest on a deposit of Rs 1000 in 1 year. The rate of interest is:

(a) 10% (b) 90% (c) 9% (d) 1%

Answer Key - Chapter 7

1. (b) 2. (b) 3. (d) 4. (c) 5. (a) 6. (d) 7. (b) 8. (a) 9. (c) 10. (a) 11. (c) 12. (b) 13. (d) 14. (d) 15. (a) 16. (c)

Try Yourself - Quick Drills

1. SI on Rs 500 at 10% for 1 year = ?

2. SI on Rs 2000 at 5% for 2 years = ?

3. P Rs 1000 aur SI Rs 150 - amount = ?

4. SI Rs 100, P Rs 500, T 2 years - rate = ?

5. SI on Rs 300 at 4% for 3 years = ?

Drill Answers: 1. Rs 50 2. Rs 200 3. Rs 1150 4. 10% 5. Rs 36

Chapter 8

Algebra Basics

Algebra mein unknown number ko letter (x, y, a) se likhte hain. Ghabrana nahi - algebra sirf hisaab ko short form mein likhna hai.

Notes - Asaan Alfaaz Mein

- **Expression:** numbers aur letters ka group, jaise 3x + 5. Jab x = 4 ho to value = 3(4) + 5 = 17.
- **Term aur coefficient:** 7x mein 7, x ka **coefficient** hai. **Like terms** woh hain jin ka letter hissa ek jaisa ho: 4xy aur 9xy like terms hain, 4x aur 4y nahi.
- Like terms jama/minus ho sakte hain: 4a + 3a = 7a, lekin 4a + 3b aise hi rahega.
- **Equation hal karna:** letter ko ek taraf akela karo. Jo operation ek taraf lago, wahi dusri taraf bhi. x + 7 = 12, to x = 12 - 7 = 5.
- Multiplication wali equation: 4x = 28, to x = 28 / 4 = 7.
- **Brackets kholna:** 5(x + 2) = 5x + 10 - bahar wala number andar dono terms se multiply hota hai.
- **Powers:** x2 ka matlab x x x hai. x = 6 ho to x2 = 36.

Worked Examples

Example 1: 3x + 4 = 19 hal karein

Pehle 4 ko dusri taraf le jayein: 3x = 19 - 4 = 15

Phir 3 se divide karein: x = 15 / 3 = 5.

Check: 3(5) + 4 = 19 - durust hai.

Example 2: a = 2 aur b = 3 ho to ab + a ki value

ab + a = (2 x 3) + 2 = 6 + 2 = 8.

Example 3: x = 3 ho to 2x2 ki value

2x2 = 2 x (3 x 3) = 2 x 9 = 18.

Aam Ghaltiyan - Common Mistakes

- 3x ka matlab 3 + x samajh lena - asal mein 3x ka matlab 3 x x hai.
- Equation mein ek taraf operation karke dusri taraf bhool jana.

Practice MCQs

1. If x = 4, the value of 3x + 5 is:

(a) 15 (b) 17 (c) 20 (d) 12

2. 4a + 3a simplifies to:

(a) 12a (b) 7a (c) 7a2 (d) 7

3. If x + 8 = 15, then x = ?

(a) 23 (b) 6 (c) 7 (d) 9

4. Solve: 4x = 28. Then x = ?

(a) 6 (b) 8 (c) 112 (d) 7

5. Solve: x - 9 = 15. Then x = ?

(a) 24 (b) 6 (c) 20 (d) 25

6. 5(x + 2) is equal to:

(a) 5x + 2 (b) x + 10 (c) 5x + 10 (d) 5x + 7

7. If x = 6, the value of x2 is:

(a) 12 (b) 30 (c) 42 (d) 36

8. Solve: 3x + 4 = 19. Then x = ?

(a) 5 (b) 6 (c) 7 (d) 4

9. The coefficient of x in 7x + 3 is:

(a) 3 (b) 7 (c) 10 (d) x

10. Which pair shows like terms?

(a) 4x and 4y (b) 3x2 and 3x (c) 5xy and 5x (d) 4xy and 9xy

11. Solve: x/3 = 6. Then x = ?

(a) 2 (b) 9 (c) 18 (d) 21

12. The sum of two numbers is 30. If one number is x, the other number is:

(a) x - 30 (b) 30 - x (c) 30x (d) x/30

13. 6y - 2y + y simplifies to:

(a) 5y (b) 4y (c) 9y (d) 3y

14. If a = 2 and b = 3, the value of ab + a is:

(a) 5 (b) 8 (c) 9 (d) 12

15. Solve: 4x - 8 = 8. Then x = ?

(a) 4 (b) 2 (c) 6 (d) 0

16. Ahmad is x years old. His age after 5 years will be:

(a) 5x (b) x - 5 (c) x/5 (d) x + 5

17. Solve: 2(x - 3) = 10. Then x = ?

(a) 5 (b) 6 (c) 8 (d) 13

18. If p = 3, the value of 10 - 2p is:

(a) 4 (b) 6 (c) 14 (d) 24

Answer Key - Chapter 8

1. (b) 2. (b) 3. (c) 4. (d) 5. (a) 6. (c) 7. (d) 8. (a) 9. (b) 10. (d) 11. (c) 12. (b) 13. (a) 14. (b) 15. (a) 16. (d) 17. (c) 18. (a)

Try Yourself - Quick Drills

1. x + 4 = 9, to x = ?
2. 6a - 2a = ?
3. y = 2 ho to 5y = ?
4. 3(x + 1) ko khol kar likhein
5. x/5 = 4, to x = ?

Drill Answers: 1. 5 2. 4a 3. 10 4. 3x + 3 5. 20

Chapter 9

Geometry Basics

Geometry shaklon (shapes) ki science hai - lines, angles aur figures. Is chapter mein angles, triangles aur area/perimeter ke bunyadi sawal hain.

Notes - Asaan Alfaaz Mein

- **Angles:** Acute angle 90 degrees se chhota, Right angle bilkul 90 degrees, Obtuse angle 90 se bara lekin 180 se chhota, Straight angle bilkul 180 degrees.
- **Complementary angles** ka sum 90 degrees hota hai. **Supplementary angles** ka sum 180 degrees hota hai.
- **Triangle ke teeno angles ka sum hamesha 180 degrees hota hai.**
- **Triangles ki types:** Equilateral (teeno sides barabar), Isosceles (do sides barabar), Scalene (koi side barabar nahi).
- **Perimeter** = tamam sides ka sum (ghera). Square ka perimeter = 4 x side. Rectangle ka = 2 x (length + width).
- **Area:** Square = side x side. Rectangle = length x width. Triangle = 1/2 x base x height. Circle = pi x r x r (pi = 22/7).
- **Circle ki circumference** (ghera) = 2 x pi x r. **Pythagoras rule:** right triangle mein hypotenuse2 = base2 + height2. 3-4-5 wala mashhoor group yaad rakhein.

Worked Examples

Example 1: Ek triangle ke do angles 60 aur 70 degrees hain - teesra angle?

Teeno angles ka sum = 180 degrees

Teesra angle = 180 - 60 - 70 = 50 degrees.

Example 2: Rectangle ki length 8 cm aur width 5 cm hai - area aur perimeter?

Area = 8 x 5 = 40 cm2

Perimeter = 2 x (8 + 5) = 26 cm.

Example 3: Radius 7 cm wale circle ka area

Area = pi x r x r = 22/7 x 7 x 7

= 154 cm2.

Aam Ghaltiyan - Common Mistakes

- Area ke jawab mein cm aur perimeter ke jawab mein cm2 likh dena - units ulta ho jate hain.
- Triangle ke angles ka sum 90 samajh lena - asal sum 180 degrees hai.

Practice MCQs

1. A right angle measures:

(a) 45 degrees (b) 180 degrees (c) 60 degrees (d) 90 degrees

2. Which of the following is an acute angle?

(a) 95 degrees (b) 120 degrees (c) 35 degrees (d) 180 degrees

3. Which of the following is an obtuse angle?

(a) 80 degrees (b) 100 degrees (c) 90 degrees (d) 45 degrees

4. The sum of the angles of a triangle is:

(a) 90 degrees (b) 360 degrees (c) 180 degrees (d) 270 degrees

5. A triangle with all three sides equal is called:

(a) Isosceles (b) Equilateral (c) Scalene (d) Right-angled

6. Two angles of a triangle are 60 and 70 degrees. The third angle is:

(a) 50 degrees (b) 40 degrees (c) 60 degrees (d) 130 degrees

7. Complementary angles have a sum of:

(a) 180 degrees (b) 360 degrees (c) 45 degrees (d) 90 degrees

8. Supplementary angles have a sum of:

(a) 180 degrees (b) 90 degrees (c) 270 degrees (d) 360 degrees

9. The perimeter of a square of side 6 cm is:

(a) 36 cm (b) 24 cm (c) 12 cm (d) 18 cm

10. The area of a rectangle 8 cm long and 5 cm wide is:

(a) 26 cm2 (b) 13 cm2 (c) 40 cm2 (d) 45 cm2

11. The area of a square of side 9 cm is:

(a) 18 cm2 (b) 36 cm2 (c) 90 cm2 (d) 81 cm2

12. The perimeter of a rectangle with length 10 cm and width 4 cm is:

(a) 40 cm (b) 14 cm (c) 28 cm (d) 24 cm

13. In a right triangle, the two shorter sides are 3 cm and 4 cm. The hypotenuse is:

(a) 5 cm (b) 6 cm (c) 7 cm (d) 25 cm

14. The area of a triangle with base 10 cm and height 6 cm is:

(a) 60 cm2 (b) 16 cm2 (c) 36 cm2 (d) 30 cm2

15. A straight angle measures:

(a) 90 degrees (b) 180 degrees (c) 120 degrees (d) 360 degrees

16. Using pi = 22/7, the circumference of a circle of radius 7 cm is:

(a) 22 cm (b) 44 cm (c) 154 cm (d) 49 cm

17. Two angles on a straight line are supplementary. If one angle is 110 degrees, the other is:

(a) 80 degrees (b) 90 degrees (c) 290 degrees (d) 70 degrees

18. An isosceles triangle has:

(a) All three sides equal (b) No side equal (c) Two equal sides (d) Four equal sides

Answer Key - Chapter 9

1. (d) 2. (c) 3. (b) 4. (c) 5. (b) 6. (a) 7. (d) 8. (a) 9. (b) 10. (c) 11. (d) 12. (c) 13. (a) 14. (d) 15. (b) 16. (b) 17. (d) 18. (c)

Try Yourself - Quick Drills

1. Square ki side 5 cm - perimeter = ?
2. Triangle ke angles ka sum = ?
3. Rectangle 6 cm x 2 cm ka area = ?
4. 30 degrees ka complementary angle = ?
5. Straight angle = ?

Drill Answers: 1. 20 cm 2. 180 degrees 3. 12 cm2 4. 60 degrees 5. 180 degrees

Chapter 10

Time, Speed & Work

Safar (travel) aur kaam (work) ke sawal ek hi soch par chalte hain: kul cheez ko kul waqt se joro. Formula seedha hai, bas unit ka khayal rakhein.

Notes - Asaan Alfaaz Mein

- **Speed = Distance / Time.** Isi se: Distance = Speed x Time, aur Time = Distance / Speed.
- **Unit badalna:** km/h ko m/s mein badalne ke liye 5/18 se multiply karein (36 km/h = 10 m/s). m/s ko km/h banane ke liye 18/5 se multiply karein.
- **Average speed = Total distance / Total time** - do speeds ka seedha average kabhi na lein.
- **Work ka usool:** agar A ek kaam 6 din mein karta hai to ek din ka kaam = 1/6 hissa.
- Do log mil kar: (A ka ek din ka kaam) + (B ka ek din ka kaam) = dono ka ek din ka kaam. Phir din = 1 / (ek din ka kaam).
- **Men-days rule:** kaam fix ho to men x days barabar rehta hai. Aadmi dugne, din aadhe.

Worked Examples

Example 1: Ek car 120 km ka safar 2 hours mein karti hai - speed?

Speed = Distance / Time = 120 / 2 = 60 km/h.

Example 2: A ek kaam 4 din mein aur B bhi wahi kaam 4 din mein karta hai - dono mil kar kitne din?

Ek din ka kaam: A = 1/4, B = 1/4. Dono mil kar = 1/4 + 1/4 = 1/2.

Kaam mukammal hone mein = 1 / (1/2) = 2 din.

Example 3: 90 km ka safar 30 km/h ki speed se - waqt?

Time = Distance / Speed = 90 / 30 = 3 hours.

Aam Ghaltiyan - Common Mistakes

- km/h aur m/s ko mix kar dena - pehle unit theek karein.
- Work wale sawal mein dono logon ke kaam ko minus kar dena - mil kar kaam JAMA hota hai.

Practice MCQs

1. Speed is calculated as:

(a) Distance / Time (b) Distance x Time (c) Time / Distance (d) Distance + Time

2. A car covers 120 km in 2 hours. Its speed is:

(a) 40 km/h (b) 100 km/h (c) 240 km/h (d) 60 km/h

3. A bus covers 90 km in 3 hours. Its speed is:

(a) 27 km/h (b) 30 km/h (c) 45 km/h (d) 270 km/h

4. 36 km/h is equal to:

(a) 10 m/s (b) 5 m/s (c) 18 m/s (d) 36 m/s

5. A train travels at 50 km/h for 3 hours. The distance covered is:

(a) 100 km (b) 120 km (c) 150 km (d) 200 km

6. How long will a car take to cover 200 km at 40 km/h?

(a) 5 hours (b) 4 hours (c) 8 hours (d) 10 hours

7. A man walks 2 km in 30 minutes. His speed is:

(a) 2 km/h (b) 6 km/h (c) 4 km/h (d) 1 km/h

8. A can complete a job in 6 days. The part of the job he does in one day is:

(a) 6 (b) 1/6 (c) 1/3 (d) 1/12

9. A alone can do a job in 4 days and B alone can also do it in 4 days. Working together, they will finish it in:

(a) 4 days (b) 8 days (c) 1 day (d) 2 days

10. A train travels at 60 km/h for 2.5 hours. The distance covered is:

(a) 120 km (b) 180 km (c) 240 km (d) 150 km

11. A cyclist travelling at 15 km/h will cover 45 km in:

(a) 3 hours (b) 2 hours (c) 4 hours (d) 5 hours

12. A journey of 100 km is completed in 4 hours. The average speed is:

(a) 20 km/h (b) 40 km/h (c) 25 km/h (d) 400 km/h

13. One man can finish a job in 12 days. Three men working together will finish it in:

(a) 36 days (b) 4 days (c) 6 days (d) 9 days

14. 72 km/h is equal to:

(a) 15 m/s (b) 20 m/s (c) 25 m/s (d) 72 m/s

15. A bus travels 180 km at a speed of 45 km/h. The time taken is:

(a) 3 hours (b) 5 hours (c) 4 hours (d) 6 hours

16. A pipe can fill a tank in 8 hours. The part of the tank filled in 1 hour is:

(a) 8 (b) 1/4 (c) 1/16 (d) 1/8

Answer Key - Chapter 10

1. (a) 2. (d) 3. (b) 4. (a) 5. (c) 6. (a) 7. (c) 8. (b) 9. (d) 10. (d) 11. (a) 12. (c) 13. (b) 14. (b) 15. (c) 16. (d)

Try Yourself - Quick Drills

1. 60 km safar 2 hours mein - speed = ?
2. 20 km/h ki speed se 3 hours ka safar = ?
3. 18 km/h = ? m/s
4. A ek kaam 10 din mein karta hai - ek din ka kaam = ?
5. 120 km, speed 60 km/h - waqt = ?

Drill Answers: 1. 30 km/h 2. 60 km 3. 5 m/s 4. 1/10 5. 2 hours

Final Test

Mixed Practice Test

Ye test tamam chapters se mila jula hai - bilkul asal test ki tarah. Waqt note karein aur khud ko check karein.

1. What is the LCM of 6 and 8?

(a) 24 (b) 12 (c) 48 (d) 14

2. 15% of 200 is:

(a) 15 (b) 45 (c) 30 (d) 3

3. The average of 5, 15 and 25 is:

(a) 10 (b) 20 (c) 45 (d) 15

4. 3/5 + 1/5 = ?

(a) 4/5 (b) 4/10 (c) 3/25 (d) 1

5. The simple interest on Rs 1000 at 10% per year for 1 year is:

(a) Rs 10 (b) Rs 100 (c) Rs 110 (d) Rs 1000

6. If x + 9 = 20, then x = ?

(a) 29 (b) 9 (c) 15 (d) 11

7. The area of a square of side 7 cm is:

(a) 28 cm2 (b) 14 cm2 (c) 49 cm2 (d) 56 cm2

8. The ratio 10 : 15 in lowest form is:

(a) 3 : 2 (b) 2 : 3 (c) 5 : 3 (d) 1 : 5

9. An article bought for Rs 300 is sold for Rs 360. The profit percent is:

(a) 20% (b) 10% (c) 25% (d) 60%

10. 0.4 written as a percent is:

(a) 4% (b) 40% (c) 0.4% (d) 400%

11. The perimeter of a rectangle with length 12 cm and width 3 cm is:

(a) 30 cm (b) 36 cm (c) 15 cm (d) 24 cm

12. 25 x 8 = ?

(a) 180 (b) 210 (c) 225 (d) 200

13. One hour is equal to:

(a) 60 seconds (b) 600 seconds (c) 3600 seconds (d) 360 seconds

14. Two angles of a triangle are 90 and 45 degrees. The third angle is:

(a) 45 degrees (b) 60 degrees (c) 90 degrees (d) 135 degrees

15. A car travels at 80 km/h for 2 hours. The distance covered is:

(a) 120 km (b) 180 km (c) 82 km (d) 160 km

16. Half of 3/4 is:

(a) 3/2 (b) 6/4 (c) 3/8 (d) 1/2

17. What is the HCF of 9 and 15?

(a) 45 (b) 3 (c) 5 (d) 6

18. 0.125 written as a fraction is:

(a) 1/4 (b) 1/6 (c) 1/8 (d) 1/10

19. 60% of 50 is:

(a) 20 (b) 30 (c) 35 (d) 60

20. The average of 3, 6, 9 and 12 is:

(a) 7.5 (b) 7 (c) 8 (d) 30

21. The ratio 18 : 24 in lowest form is:

(a) 4 : 3 (b) 2 : 3 (c) 3 : 5 (d) 3 : 4

22. An article bought for Rs 150 is sold for Rs 180. The profit percent is:

(a) 20% (b) 15% (c) 25% (d) 30%

23. The simple interest on Rs 400 at 5% per year for 3 years is:

(a) Rs 20 (b) Rs 60 (c) Rs 75 (d) Rs 100

24. Solve: 5x - 5 = 20. Then x = ?

(a) 3 (b) 4 (c) 5 (d) 15

25. The supplement of a 65-degree angle is:

(a) 25 degrees (b) 125 degrees (c) 295 degrees (d) 115 degrees

26. A car covers 45 km in 1.5 hours. Its speed is:

(a) 35 km/h (b) 40 km/h (c) 30 km/h (d) 60 km/h

27. 9 x 7 = ?

(a) 63 (b) 56 (c) 72 (d) 54

28. The perimeter of a square of side 11 cm is:

(a) 121 cm (b) 22 cm (c) 33 cm (d) 44 cm

29. 2/7 + 3/7 = ?

(a) 5/14 (b) 5/7 (c) 6/7 (d) 1/7

Answer Key - Mixed Practice Test

1. (a) 2. (c) 3. (d) 4. (a) 5. (b) 6. (d) 7. (c) 8. (b) 9. (a) 10. (b) 11. (a) 12. (d) 13. (c) 14. (a) 15. (d) 16. (c) 17. (b) 18. (c) 19. (b) 20. (a) 21. (d) 22. (a) 23. (b) 24. (c) 25. (d) 26. (c) 27. (a) 28. (d) 29. (b)

Word List

Maths Glossary - Alfaaz ke Matlab

Is list mein is book ke tamam aham alfaaz ke asaan matlab diye gaye hain. Jo lafz samajh na aaye, yahan se dekh lein.

- **Factor:** woh number jo kisi number ko poora-poora divide kare (baghair remainder).
- **Multiple:** kisi number ko 1, 2, 3 ... se multiply karne se bane numbers.
- **Prime Number:** jis ke sirf do factors hon - 1 aur woh khud.
- **Numerator:** fraction ka upar wala hissa.
- **Denominator:** fraction ka neeche wala hissa.
- **Reciprocal:** fraction ko ulta karna - 4/7 ka reciprocal 7/4.
- **Percent:** har 100 mein se - 40% matlab 100 mein se 40.
- **Ratio:** do cheezon ka muqabla, jaise 2 : 3.
- **Proportion:** do barabar ratios, jaise 2 : 3 = 4 : 6.
- **Average (Ost):** kul jama ko tadaad par taqseem karna.
- **Cost Price (CP):** jis qeemat par cheez khareedi jaye.
- **Selling Price (SP):** jis qeemat par cheez bechi jaye.
- **Marked Price (MP):** cheez par likhi hui qeemat.
- **Discount:** marked price par di gayi riayat (kami).
- **Principal (P):** asal rakam jis par interest banta hai.
- **Rate (R):** interest ya percent ki sharah - % per year.
- **Equation:** barabari wala jumla jismein unknown letter ho, jaise x + 5 = 9.
- **Coefficient:** letter ke sath multiply hone wala number - 7x mein 7.
- **Angle:** do lines ke milne se bane jhukao ki pemaish (degrees mein).
- **Perimeter:** kisi shape ke ghere (tamam sides) ka total.
- **Area:** kisi shape ke andar ki jagah - cm2 ya m2 mein.
- **Circumference:** circle ka ghera.
- **Hypotenuse:** right triangle ki sab se lambi side.
- **Speed:** ek ghante (ya second) mein taya hua fasla.

Revision

Golden Formula Chart

Test se pehle ye chart ek nazar zaroor dekhein - tamam chapters ke formulas ek jagah.

- Percentage of a quantity = (Percent / 100) x Quantity
- Percentage = (Part / Whole) x 100
- Percentage change = (Change / Original value) x 100
- Average = Sum of values / Number of values | Sum = Average x Number of values
- Profit = SP - CP | Loss = CP - SP | Profit% or Loss% is always on CP
- SP = CP x (100 + Profit%) / 100 | SP = CP x (100 - Loss%) / 100
- Simple Interest: SI = (P x R x T) / 100 | Amount = P + SI
- Ratio division: one part = Total amount / Total parts
- Perimeter of square = 4 x side | Perimeter of rectangle = 2 x (length + width)
- Area of square = side x side | Area of rectangle = length x width
- Area of triangle = 1/2 x base x height | Area of circle = pi x r x r (pi = 22/7)
- Circumference of circle = 2 x pi x r
- Sum of angles of a triangle = 180 degrees | Complementary = 90, Supplementary = 180
- Speed = Distance / Time | Distance = Speed x Time | Time = Distance / Speed
- km/h to m/s: multiply by 5/18 | m/s to km/h: multiply by 18/5
- One day work = 1 / (total days for the job)
- HCF x LCM = First number x Second number (for two numbers)

Exam Tips

- Pehle asaan sawal hal karein - mushkil sawal par waqt zaya na karein, nishan laga kar aage barhein.
- Har sawal ko ek dafa ghour se parhein - aadha jawab aksar sawal ke andar chhupa hota hai.
- Percentage aur profit-loss mein hamesha ORIGINAL / CP value se divide karein.
- Apna jawab options se match karke check karein - unit (cm, cm2, km/h) ka khaas khayal rakhein.
- Rough work saaf likhein - galti pakarne mein asani hoti hai.

Best of luck! Mehnat kabhi zaya nahi jati - roz ka thora sa parhna, kal ki bari kamyabi banta hai. - MR Academy